

Question Number	Answer	Additional Guidance	Mark
4(a)	A description that includes three of the following: • find a suitable pre-treatment for seeds (1) • find suitable conditions for {germination/growth/respiration} of seeds (1) • find a suitable time for seeds to (start) {germination/growth/respiring} (1) • find a suitable timescale to measure distance moved by bubble (1)	Accept soaking time / number of times to rinse seeds /other valid example Accept temperature / light wavelength/daylength Accept time to produce a measurable change of gas find a suitable {number / mass} of seeds to obtain a measurable rate/quantity of gas	Exp (3)

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4(b)	An answer that includes eight of the following: • clear statement of the dependent variable (1) • suitable method for preparing seeds (1) • use of a carbon dioxide absorber (1) • seeds left to equilibrate/acclimatise (before data collection) (1) • description of data collection in a given time (1) • calculation of rate described (1) • identify one abiotic variable to be controlled and description of how it is controlled (1) • one biotic variable identified (1) • use both types of seed in the investigation (1)	Accept distance moved by bubble (in respirometer) in given time / volume of oxygen/gas used in given time Accept soaking / rinsing / surface sterilisation etc E.g soda lime /KOH/NaOH Not sodium hydrogen carbonate Accept distance moved by bubble in given time / volume of gas collected in syringe E.g distance / volume divided by time Accept: temperature / wavelength of light Ignore pH Accept age of seeds / mass of seeds / variety of seeds	Exp (8)

Question Number	Answer	Additional Guidance	Mark
4(c)	A description including the following raw data table with headings and units, with means calculated from repeats (1) - suitable graph with labelled axes (1) - use of an appropriate statistical test (1)	Accept distance moved by bubble (in given time) Accept additional columns eg volume of oxygen used / rate of respiration Accept description of calculating mean Do not accept units in the body of the table Accept distance moved by bubble / volume of oxygen used / rate of respiration MP2 and 3 usually a bar graph and a stats test for difference OR a line graph of time intervals	Exp (3)

Question Number	Answer	Additional Guidance	Mark
4(d)	An answer that includes three of the following: • if mung beans germinate more quickly than peas the highest respiration rate may be at different times after soaking (1) • It is hard to control {age/viability}of seeds (1) • Seed may change from aerobic to anaerobic respiration (1) • errors in { reading position of bubble on scale/measuring volume of gas} (1) • germinating seeds produce heat so hard to control temperature (which affects rate of respiration) (1)	Accept older seeds may have a lower {germination rate / respiration rate} Accept if some seeds do not germinate this will affect the result / reduce the respiration rate Accept changing temperature will affect position of bubble on scale	Exp (3)

(Total for question 4 = 17 marks)

